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Katrina's Children: Evidence on the Structure of Peer Effects from Hurricane Evacuees†

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In 2005, Hurricanes Katrina and Rita forced many children to relocate across the Southeast. While schools quickly enrolled evacuees, families in receiving schools worried about the impacts on incumbent students. We find no effect, on average, of the inflow of evacuees on achievement in Houston. In Louisiana we find little impact on average and we reject linear-in-means models. Moreover, we find that student achievement improves with high achieving peers and worsens with low achieving peers. Finally, an increase in the inflow of evacuees raised incumbent absenteeism and disciplinary problems in Houston's secondary schools. (*JEL* I21, Q54)

Hurricane Katrina made landfall in southeast Louisiana on August 29, 2005 as the most destructive and costly hurricane ever in the United States, with estimated damage of over $80 billion (Knabb, Rhome, and Brown 2005). Eighty percent of New Orleans as well as large parts of Mississippi and Alabama were flooded. Just a few weeks later, Hurricane Rita hit Louisiana and east Texas as the most powerful storm ever recorded in the Gulf. While Rita hit a less populated area, it nonetheless caused substantial damage.

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Hurricanes Katrina and Rita forced over one million people to evacuate, generating one of the greatest rapid migrations of children and their families in US history (Ladd, Marszalek, and Gill 2006). Some areas of Louisiana received large numbers of evacuees. Baton Rouge received about 200,000 evacuees and Hammond received over 10,000, nearly doubling their populations.[1] Many evacuees left the affected states. Houston, Texas received 75,000, the largest number of evacuees received by any city outside Louisiana (McIntosh 2008).

As a result of the migration, many children were uprooted. School districts mounted substantial efforts to enroll the evacuees as quickly as possible. While Baton Rouge, Houston, and other cities that took in large numbers of evacuees were seen as examples of solidarity, the inflow of new students into schools created concerns among the nonevacuee population. Evacuee children came from some of the worst-performing schools in the country and parents worried that their children would be negatively affected by the inflow of poor performing peers.

In this paper, we use administrative data from the Houston Independent School District (HISD) and the Louisiana Department of Education to examine whether the inflow of Katrina and Rita students adversely affected the academic performance, attendance, and discipline of their new peers.[2] Much of the literature on peer effects for higher education finds modest positive peer effects on grade point average (e.g., Carrell, Fullerton, and West 2009; Lyle 2007; Sacerdote 2001; Stinebrickner and Stinebrickner 2006; and Zimmerman 2003), but results for elementary and secondary education are more mixed with some studies finding little or no effects (e.g., Angrist and Lang 2004; Hanushek et al. 2003; Vigdor and Nechyba 2007) and others finding large effects (e.g., Hoxby 2000; Hoxby and Weingarth 2006; Lavy and Schlosser 2007; Lavy, Paserman, and Schlosser 2008). Research has also shown that poor peer behavior worsens children's own behavior and achievement (e.g., Carrell and Hoekstra 2010; Carrell, Malmstrom, and West 2008; Figlio 2007; Gould, Lavy, and Paserman 2009; Kling, Leibman, and Katz 2007; Lavy and Schlosser 2007).

This paper stands out from the growing literature on education peer effects in a number of ways. First, the magnitude of the inflow of evacuees in Houston and Louisiana provides much larger variation in the composition of peers than comparable studies that exploit year-to-year variation within schools. It is hard to think of any event that caused as large, sudden, and arguably exogenous a change in peer effects as Hurricanes Katrina and Rita. Admittedly, this large shift in peer composition does come with a trade-off; the hurricanes may also have had effects that work through other channels such as teacher and school-level responses. While we do our best to rule out effects that operate through class size or other resource constraints, we cannot be completely certain that our results extend to typically observed levels of variation in peer quality.

[1] According to our data, while 93 percent of eventual student evacuees resided in the parishes (counties) most affected by Hurricane Katrina (Orleans, Jefferson, Plaquemines, and Saint Bernard) in 2003–2004 and 2004–2005, only 68 percent resided in those parishes in 2005–2006, increasing to 76 percent in 2006–2007.

[2] Other work has considered the impact of Hurricanes Katrina and Rita on evacuees. Sacerdote (2008) shows increased test scores for evacuees in the medium-term from attending new schools. This study is similar to Gould, Lavy, and Paserman (2004), which examines the impact of access to better schools in Israel for children evacuated from Ethiopia. Vigdor (2008), Groen and Polivka (2008), and Belasen and Polachek (2008) look at the economic impacts of the hurricanes on New Orleans and evacuee labor supply. Paxson and Rouse (2008) look at what caused evacuees to return to New Orleans.

Second, since we have panel data on students and schools that allow us to control for prior student performance and school fixed-effects, our identification relies on the exogeneity of the inflow of evacuees with respect to *changes* in peer quality in a school. Given the unexpected nature of these hurricanes, haphazard evacuation process, and chaotic arrival to their destinations, students were unlikely to select into schools based on preexisting trends. Indeed, our placebo tests show little evidence that evacuees tended to go to schools with changing student performance.[3] Thus, our identification overcomes the usual problems of selection and reflection, and the correlated unobservables problem in the estimation of peer effects.[4]

Third, most of the literature in this area focuses on the existence of peer effects, but few studies say anything about their functional forms and underlying causes, with the exceptions of the experiment in Kenya in Duflo, Dupas, and Kremer (2011); Lavy, Paserman, and Schlosser (2008) in Israel; and Hoxby and Weingarth (2006) for the United States, which tests different models of peer effects. Here too, we focus on the nature of peer-to-peer interactions, by testing whether better peers improve performance and worse peers lower it in a monotonic fashion, or whether nonmonotonicities exist due to, for example, benefits from ability grouping. Finally, our paper relates to the literature on the impact of peers on outcomes beyond academic performance. We consider the effect of peers on disciplinary infractions and absenteeism in schools, using administrative rather than self-reported data used in other studies.

We find that, on average, the inflow of evacuees had little impact on incumbent students' achievement in Louisiana. We also find no mean effect on achievement in Houston, although we do find evidence of worsening attendance and behavior. We also estimate standard linear-in-means models for achievement using the Katrina/Rita shares and their contribution to mean peer achievement to instrument for the average performance of peers and test whether the linear-in-means assumption holds, rejecting such a model for middle/high school students.

We find evidence for monotonicity in peer effects; i.e., that all students benefit from high-achieving peers and are hurt by low-achieving peers and reject invidious comparison and ability grouping models, where grouping is done at the school-grade level. For these results to be driven by preexisting trends, poor-performing evacuees in a given grade would have had to attend generally declining schools and high-performing evacuees in a given grade would have had to enroll in generally improving schools. Given the chaotic nature of the evacuation, such a selection story seems implausible. Nonetheless, we try to dispel concerns about selection on preexisting trends or on changes in performance prior to the hurricanes by including controls for school-specific and school-type time trends, placebo tests, and a wide variety of specification and validity tests, which we will outline later in detail. In addition, our results do not appear to be driven by disruption from entry and exit of evacuees. Finally, we examine potential avenues other than peer effects through which evacuees may

[3] We further check this assumption by adding school-grade fixed effects to some specifications. In this case, the identification relies on a weaker assumption that students do not sort into grades within schools on trends. The results are qualitatively similar and are provided in the online Appendix.

[4] See Manski (1993) and Moffitt (2001) for detailed explanations of the problems that arise in the estimation of peer effects.

affect incumbent students. In particular, we consider the impact of the Katrina/Rita shares on class sizes, per-student expenditures, and average teacher experience. These results show no statistically significant impacts, with the exception that in Houston elementary schools teacher credentials *increase* slightly. In Houston, we also test whether incumbent student mobility responded to the inflow of evacuees and find little impact overall.[5]

I. Hurricane Katrina: A Natural Experiment from a Natural Disaster

A. *Assignment to Schools*

In 2005 Hurricanes Katrina and Rita caused severe damage, forcing many schools to close for an extended period of time. As a result, 188,000 public school students were forced to enroll in new schools.[6] Fortunately, school districts across the country acted quickly to open their doors to evacuated students. In Louisiana 56 districts not damaged by the hurricanes enrolled about 22,000 displaced students, with the rest enrolling in nearly every state in the United States, with a majority of 45,000 going to Texas.

Within Louisiana, many people evacuated to places where they had family and friends, but many others ended up in shelters and hotels. Of the 200,000 evacuees in Baton Rouge, about 10,000 were living in shelters. Evacuees in East Baton Rouge mainly subsisted on Federal Emergency Management Agency assistance and resided in inexpensive hotels and apartments (Tanneeru 2005). A superintendent of a school district in Baton Rouge told us that "every hotel and motel in the parish was completely full and that almost everyone in the parish had someone in their house who they either knew directly or knew through someone else."[7] The housing situation in Houston was similar to that in Baton Rouge. On August 31, Reliant Astrodome received the first bus of evacuees coming from the New Orleans Superdome and it ended up housing 30,400 evacuees in total. Many more evacuees were housed in the George R. Brown Convention Center and in 36 Red Cross shelters throughout the city, with others staying in motels or cheap apartments.

Given that evacuees' lives were in flux, schools were not the primary concern for many parents. In an interview, Tami Head, a mother of four in East Baton Rouge, indicated that she was "too overwhelmed to think much about school" (Tanneeru 2005). For the most part, parents were hardly placed in a situation to think about school choices, or for that matter to be picky about any choices actually made.[8]

Nonetheless, school districts made proactive efforts to enroll children quickly. Administrators in Louisiana and Houston confirmed that students were placed

[5] Since our data for Louisiana only includes fourth-, eighth-, and tenth-grade students prior to Katrina, we cannot conduct the same analysis there.

[6] A longer discussion of the evacuation and student assignment following the hurricanes can be found in the online Appendix.

[7] Personal communication with the Superintendent of East Baton Rouge Parish School District, October 2009.

[8] It is possible that some evacuees' parents cared a great deal about education and were in a situation to spend the time and effort necessary to make the best school arrangements for their children. These are likely to have been a minority of those attending public schools, however, so any bias introduced below will be mitigated. Moreover, even if they were informed, it is unlikely that they would have been so well informed as to self-select on the basis of preexisting trends in a school which is what would be required for our results to be biased given that we exploit changes in the share of evacuees in a given school grade over time.

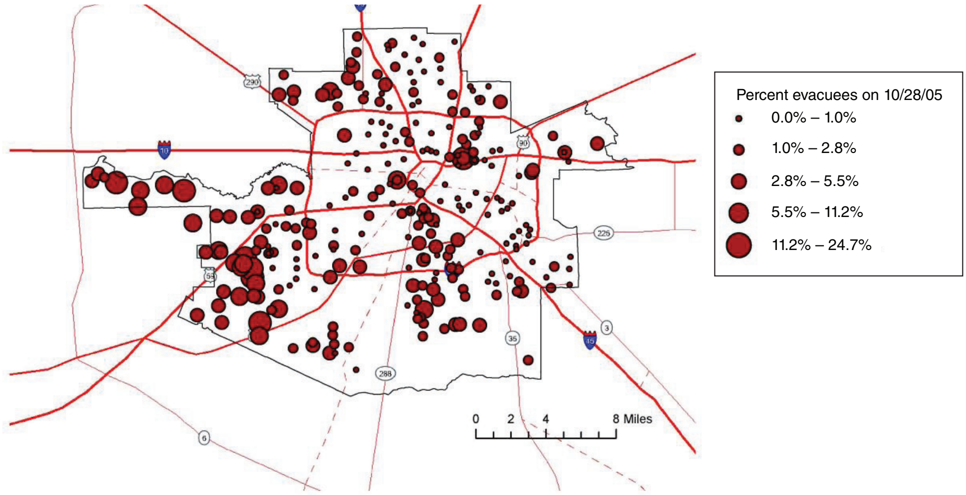

Figure 1. Hurricane Katrina and Rita Evacuees in HISD by School, 2005–2006

in schools based on their location of residence at the time. In fact, because the McKinney-Vento Homeless Education Assistance Act of 2001 requires states to enroll homeless students, schools were forced to promptly enroll the evacuees. Once enrolled, the McKinney-Vento Act also prohibits segregating homeless students from the general population. Hence, evacuees had to be included in classes with nonevacuee students.[9] Even if schools wanted to separate students by academic level, they had little information on the students upon enrollment. According to a superintendent in Baton Rouge, most paperwork was unavailable or destroyed in the flooding, so placement was based on parents' verbal information. Due to the unusual circumstances, both Texas and Louisiana waived requirements on immunization and academic records for evacuees.[10]

Figure 1 displays a map of the percentage of Katrina and Rita student evacuees in HISD schools, which shows substantial variation in terms of exposure to the evacuee children. Some schools in Houston received no evacuees at all, while the mean and maximum shares were 2.5 percent and 25 percent of students by late October, 2005. Figure 2 shows a similar map for Louisiana. Like Houston, some schools in Louisiana received no evacuees at all, while the average and largest evacuee enrollments in our analysis sample, which excludes schools in the areas directly affected by the storms and schools with more than 70 percent evacuees, were 3.1 percent and 56 percent, respectively.

It is important to point out that our empirical analysis below does not rely on the cross-sectional variation in the share of evacuees in Houston and Louisiana shown

[9] Figures 1 and 2 in the online Appendix show that evacuees in Houston were distributed widely across classrooms. While we do not have classroom data for students in Louisiana, we are able to identify the teacher who administered the standardized tests as a proxy for classroom or homeroom teacher. Figure 3 in the online Appendix shows that in Louisiana evacuees were dispersed widely across test administrators as well.

[10] The Texas Education Agency (TEA) website states that "standard immunization requirements for children attending school in Texas are being temporarily waived for children displaced by Hurricane Katrina," and the same was true of student records. http://texaslawhelp.org/tx/showdocument.cfm/doctype/dynamicdoc/ichannelprofileid/27884/idynamicdocid/3109/iChannelID/91/foreignlanguage/1.

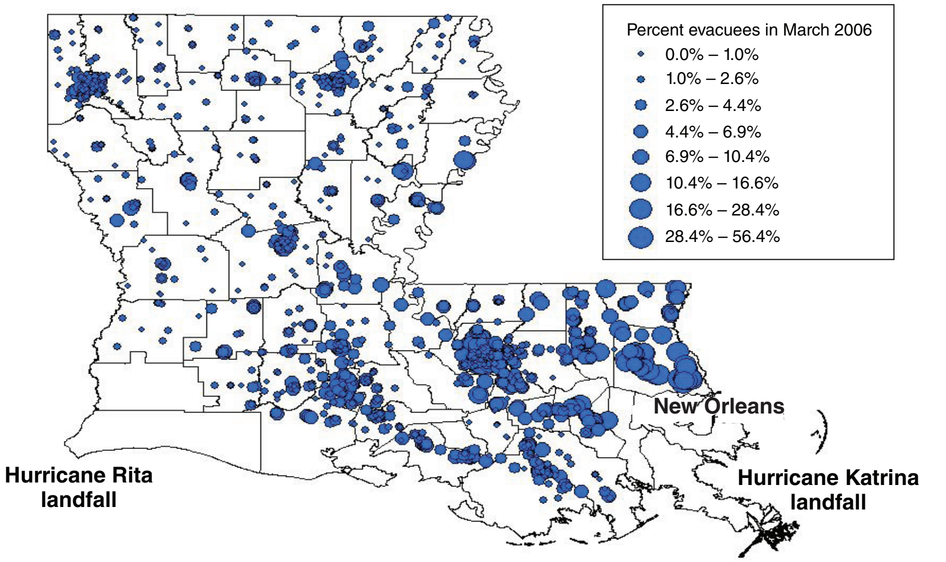

FIGURE 2. HURRICANES KATRINA AND RITA EVACUEES IN LOUISIANA BY SCHOOL, 2005–2006

Notes: Map shows Katrina and Rita evacuees as of March 2006 Louisiana Educational Assessment Program (LEAP) testing. Schools with more than 70 percent evacuees and those in Cameron and Calcasieu Parishes (Rita landfall area) and Orleans, St. Bernard, Placquemines, and Jefferson Parishes (Katrina landfall area) are excluded as these schools were likely directly affected by the storms.

in these two maps. Instead, we control for past student performance and school effects so that we account for the possibility that evacuees may be concentrated in certain neighborhoods and potentially in schools with students of different quality. Moreover, we exploit the influx of evacuees in different grades. This means that our identification is coming from the exogeneity of the inflow of evacuees in various grades to changes in school characteristics. Below, we provide empirical support for this assumption.

B. *School Responses*

While receiving districts made great efforts to accommodate the new students, many parents worried about financial burdens and overcrowding. Indeed, a superintendent in Baton Rouge described this time to us as a "very chaotic time … with everything changing from day to day." Having sufficient resources was a concern for many schools. For example, one news report said that HISD would face an extra \$20 million in costs over the 2005–2006 school year (Klein 2006). Districts were eligible for grants via the McKinney-Vento Act, however, and in 2005 Congress passed legislation to aid school districts with evacuees. This assistance ultimately amounted to \$6,000 per student (Radcliffe 2006). This, combined with temporary exemptions of evacuees from counting toward accountability requirements in Texas, ensured that schools were careful to correctly identify the evacuees in their records.[11]

[11] In Texas, evacuee performance was excluded from accountability calculations for all schools in 2005–2006, but was included starting in 2006–2007. Louisiana exempted all schools from accountability in 2005–2006.

With regards to reduced resources, ultimately schools received funding for evacuees and, as we show below, there appeared to be little impact on per-student expenditures. Another concern, however, was that schools receiving many evacuees would experience a sharp increase in class sizes. This was mitigated by class size requirements in Louisiana for all grades and in Texas for grades K–4.[12] In both states, districts with evacuees were granted temporary waivers of the class-size limits. In personal communications, however, officials in Houston and Baton Rouge indicated that these waivers were temporary, generally lasting for less than a month. Louisiana also slightly relaxed their rules but only in 2005–2006.[13] Given these requirements, it is not surprising that we find no statistically significant increase in class sizes. This means that new classes must have been created and new teachers hired. Evacuees were generally not segregated in the new classes, however. As discussed in footnote 8 above, Figures 1–3 in the online Appendix show that evacuees were broadly mixed into classrooms with incumbent students.

II. Identification Strategy

We begin by estimating a reduced-form specification of the direct impact of the inflow of Katrina and Rita evacuees into Houston and Louisiana schools on incumbent students as follows:

$$(1)\quad Y_{ig(t)jt} = \alpha + \beta Katrina\_Fraction_{jg(t)t} + \sum_{\tau=t-1}^{t-\mathrm{T}} [\delta_\tau Y_{ig(t-\tau)jt-\tau} \times LagYear_\tau] + \Omega X_{ig(t)jt} + \Pi Grade_{g(t)} + \Gamma Year_t + \Phi Grade_{g(t)} \times Year_t + \kappa_j + \varepsilon_{ig(t)jt},$$

where $Y_{ig(t)jt}$ is the academic or behavioral outcome of incumbent student i in grade g in year (t) attending school j at time t,[14] $Katrina\_Fraction_{jg(t)t}$ is the number of evacuees divided by the total number of students in school j and grade g which the student attends in year t, where this fraction is zero before the 2005–2006 academic year. The school-year t is determined by the observation date for demographic data, which is in October of calendar year t for Houston, and March of year $t+1$ for Louisiana; $Y_{ig(t-\tau)jt-\tau}$ is the lagged outcome for individual i in the last pre-Katrina year, $t-\tau$, in the grade the student attended in year $t-\tau$, for which the student has information available. This is interacted with an indicator variable for the number of years since the lagged exam was taken, $LagYear_\tau$, to account for

[12] For Louisiana, maximum enrollment is 26 students/class for grades K–3 and 33 students for grades 4–12. In Houston, the requirement is a maximum of 22 students/class in grades K–4.

[13] The requirement in Louisiana was temporarily increased to 28 students in grades K–3 and to 35 students in grades 4–12.

[14] While some peer effect studies examine the effect of peer group changes on both the receiving and the new students, we feel that peer effect studies focusing on the incumbent students are much stronger as the incoming students are not only grouped with new students but are also moved, bused, and in our case affected by the hurricanes, which is a different treatment than just having different peers.

multiple years of growth.[15] This value-added specification allows us to focus on the impact of the inflow of Katrina evacuees on changes in performance relative to a baseline; $X_{ig(t)jt}$ are observable characteristics of the student, including indicators for being female, white, African American, Hispanic, Asian, Native American, and qualifying for free-lunch, reduced-priced lunch, or being classified as otherwise economically disadvantaged;[16] $Grade_{g(t)}$ and $Year_t$ are grade and year effects and κ_j are school-fixed effects. Including κ_j controls for time-invariant differences in school quality for schools with differing shares of evacuees.

The coefficient β represents peer effects stemming from the share of students in one's own grade and school who are evacuees. In the terminology of Manski (1993) these are exogenous peer effects since they stem from peers' backgrounds. In order for β to represent a causal impact of the evacuees on test score growth, we need $Katrina\_Fraction_{jg(t)t}$ to be uncorrelated with unobserved school-grade characteristics after controlling for past test scores, demographic characteristics, and fixed unobservable school characteristics. Hence, we do not need the share of evacuees to be uncorrelated with school characteristics and the baseline achievement of incumbent students. Rather, we need only for evacuee shares in a grade to be uncorrelated with *changes* in school quality and incumbent performance that occur for reasons other than the arrival of the evacuees. We show below that the data support this assumption.

Given the chaos and uncertainty facing the evacuees, the initial assignment to schools was plausibly exogenous[17] to changes in school quality, so we interpret the coefficient on *Katrina_Fraction* as capturing the causal effect of the inflow of Katrina and Rita evacuees on incumbent students. After a few months, however, evacuees in temporary housing likely moved to apartment complexes and more permanent residences and may have also moved schools. In Houston, families relocated directly to: (i) more "permanent" housing (i.e., apartments or motels); (ii) "temporary" Red Cross shelters; or (iii) Reliant Stadium or the Convention Center. We know that virtually all of the Reliant Stadium and Convention Center residents had to leave the complex and change schools within two months of Katrina. Many evacuees in more "permanent" housing and those in "temporary" shelters, however, stayed in the same schools in which they initially entered. To address eventual self-selection into new schools, we provide specifications for Houston that focus on those most likely to be exogenously assigned to schools; i.e., those in groups (i) and (ii). Thus, we instrument for the end-of-year share with the share of evacuees living in shelters (except those in Reliant Stadium and the Convention Center), and with the initial fraction of Katrina evacuees in a school on September 13, 2005.[18]

[15] Results without the interaction are similar and are provided in the online Appendix.

[16] The other economic disadvantage and Native American categories are only available for HISD. Other economic disadvantage equals one for someone living in a family: (a) with an income level below or at the federal poverty line; (b) eligible for Temporary Assistance for Needy Families (TANF) or other public assistance; (c) receiving a Pell Grant or comparable state program for need-based financial assistance; (d) eligible for programs under Title II of the Job Training Partnership Act (JTPA); and/or (e) eligible for food stamps under the Food Stamp Act of 1977 and not eligible for free or reduced-price lunch. In Louisiana, free and reduced price lunch are combined into one indicator.

[17] Here we use "exogenous" in the statistical sense of the word not the Manski typology.

[18] Unfortunately, we do not have similar data on September 2005 assignments for Louisiana.

Instrumental variables results using these sources of variation are similar to our baseline model and are provided in the online Appendix.

We further test our identification by performing a placebo experiment in which we regress pre-Katrina outcomes on the future share of evacuees in a grade and school. Moreover, we check whether high- and low-performing evacuees based on pre-Katrina test scores were attracted to schools with changing quality using school level data going back to 1999.

In addition, using Houston data, we test for possible nonrandom migration of incumbent students after the arrival of the evacuees by running a specification similar to equation (1) for the probability of moving to other schools within a district and for the probability of moving to another district. Moreover, we estimate instrumental variables regressions in which the Katrina share in the incumbent's current school is instrumented with the fraction of Katrina evacuees in the school attended pre-Katrina. We also test whether changes in performance are related to the inflow of Katrina evacuees through channels other than peer effects by estimating a specification similar to equation (1) for class size, per-student expenditures, and teacher characteristics.

While peer effects can result through student interactions or teacher responses, another possibility is that the entry of new students causes disruptions. We test for this by estimating a regression of outcomes on the weekly share of evacuees, and the standard deviation of weekly enrollment instrumented with the standard deviation of weekly enrollment by evacuees. If pure disruption affected incumbent students' performance and behavior, we would expect the weekly share effect to disappear when the standard deviation is added. We do not find evidence that our peer effects are purely driven by disruption.

Throughout the paper, we focus on results based on the share of evacuees within grade and school. We have, however, estimated equation (1) using schoolwide shares as well. Schoolwide shares should give bigger effects as these capture not only the teacher responses and direct peer effects, but also changes to the school climate more generally. Estimates based on school-level shares are reported in an online Appendix and, as discussed below, give qualitatively similar results but, as expected, are bigger in magnitude.[19] The results based on the school-level shares are most relevant for middle/high school, since secondary students are more likely to be exposed to evacuees outside their grade as many classes have students from multiple grades.[20]

We then proceed to estimate standard linear-in-means models of peer effects as follows:

$$(2)\quad Y_{ig(t)jt} = \alpha + \rho Y_{-ig(t)jt} + \sum_{\tau=t-1}^{t-\mathrm{T}} [\delta_\tau Y_{ig(t-\tau)jt-\tau} \times LagYear_\tau] + \Omega X_{ig(t)jt}$$
$$+ \Pi Grade_{g(t)} + \Gamma Year_t + \Phi Grade_{g(t)} \times Year_t + \kappa_j + \varepsilon_{ig(t)jt},$$

[19] We have also estimated equation (1) using the classroom level shares in Houston and the test administrator in Louisiana (since we do not have information on the classroom). While we report results from the classroom and test administrator–level regressions in the online Appendix, we are concerned about endogeneity of placements at the classroom level by principals and teachers.

[20] While for Houston we can estimate value-added models for elementary and middle/high school, we focus our analysis on middle/high for Louisiana because we only have lagged test scores for those in higher grades as only fourth, eighth, and tenth grades were tested prior to Katrina.

where $Y_{-ig(t)jt}$ is the contemporaneous average peer test score in school j and grade g for individual i at time t. Our peer effect coefficient ρ includes both exogenous and endogenous peer effects, meaning effects that stem from peers' backgrounds and from their current outcomes (Manski 1993). As discussed above, and widely in the literature, the peer effect estimate of ρ is likely biased because of selection into schools and because of the reflection problem. We eliminate these biases by instrumenting for the average peer score with the fraction of Katrina evacuee students in the school and grade. Moreover, in Louisiana we have information about previous performance of evacuees, so we multiply the evacuee share by the mean lagged achievement of the evacuees to generate an achievement-adjusted evacuee share. Thus, the identifying assumption is that, after controlling for previous performance and demographic characteristics of the student and fixed school unobserved effects, the achievement-adjusted evacuee share is uncorrelated with current performance of the incumbent student other than through peer effects. We also examine whether peer effects differ by achievement levels of incumbents by expanding equation (2) as follows:

$$(3)\quad Y_{ig(t)jt} = \alpha + \rho_1 Y_{-ig(t)jt} \times D_{Q1it-\tau} + \rho_2 Y_{-ig(t)jt} \times D_{Q2it-\tau} + \rho_3 Y_{-ig(t)jt} \times D_{Q3it-\tau} + \rho_4 Y_{-ig(t)jt} \times D_{Q4it-\tau} + \sum_{q=1}^{4}\left[\sum_{\tau=t-1}^{t-T}[\delta\tau_q Y_{ig(t-\tau)jt-\tau q} \times LagYear_\tau] + \Omega_q X_{ig(t)jtq} + \Pi_q Grade_{g(t)q} + \Gamma_q Year_{tq} + \Phi_q Grade_{g(t)q} \times Year_t + \kappa_{jq}\right] + \varepsilon_{ig(t)jt},$$

where $D_{kit-\tau}$ is an indicator of whether incumbent student i's pre-Katrina achievement is in quartile $k = 1, 2, 3, 4$ of the nonevacuee distribution for year t.[21] The expression inside the summation sign indicates that we allow for interactions between a student's own quartile and all of the other covariates.[22] The instruments are the interactions of the evacuee share or performance weighted evacuee share with the incumbents' quartile. A necessary condition of the linear-in-means model is that the peer-effect is the same regardless of a student's position in the achievement distribution; i.e., $\rho_1 = \rho_2 = \rho_3 = \rho_4$.

Since our results reject this test for middle/high school students in Louisiana, we proceed to estimate nonlinear models. We follow an approach similar to Hoxby and Weingarth (2006).[23] First, we classify incumbent and evacuee students by their pre-Katrina achievement quartiles. Then, we estimate separate regressions for

[21] Since our pre-Katrina data in Louisiana is limited to grades four, eight, and ten in 2000–2001 through 2004–2005 and grades four and eight in 1999–2000, we use pre-Katrina test scores for the last of those years the student is observed to identify the student's quartile. Quartiles in Houston were constructed similarly using 2002–2003 through 2004–2005.

[22] We also estimate this model without interactions of lagged test scores with the indicator for the quartile in the previous period. The main specification is more flexible in that it allows the linear effect of the lagged test score on the current test score to be different depending on whether the person is in the first, second, third, or fourth quartiles.

[23] We do not have pre-Katrina achievement for Houston evacuees, thus we only conduct this analysis in Louisiana.

incumbent students in each quartile of achievement on the percentages of evacuees in their school and grade who fall in each quartile as follows:

$$
\begin{aligned}
(4)\quad E(Y_{ig(t)jt} | Q_{kt-\tau}) = \alpha &+ \beta_{q1} Katrina\_Fraction_{Q1g(t)jt-\tau} \\
&+ \beta_{q2} Katrina\_Fraction_{Q2g(t)jt-\tau} \\
&+ \beta_{q3} Katrina\_Fraction_{Q3g(t)jt-\tau} \\
&+ \beta_{q4} Katrina\_Fraction_{Q4g(t)jt-\tau} + \Omega X_{ig(t)jt} + \Pi Grade_{g(t)} \\
&+ \Gamma Year_t + \Phi Grade_{g(t)} \times Year_t + \kappa_j + \varepsilon_{ig(t)jt},
\end{aligned}
$$

where $Q_{qt-\tau}$ is the incumbent student's pre-Katrina achievement quartile q.

We test three models of peer effects.[24] The monotonicity model posits that peer effects are weakly increasing in peer quality. We conduct two tests of monotonicity. A weaker version tests whether $\beta_{k4} > \beta_{k1}$,where k is the incumbent's quartile. Our strong test requires that each quartile's estimate is greater than the estimate for the immediately preceding quartile; $\beta_{k4} > \beta_{k3}$, $\beta_{k3} > \beta_{k2}$, and $\beta_{k2} > \beta_{k1}$. In contrast, the invidious comparison model posits that a student's performance is harmed by having peers who are more able and helped by having peers who are less able. This model does not say anything about the impact of peers at the same ability level. Thus, our test of the invidious comparison model requires us to test $\beta_{kj} < 0$ ($\beta_{kj} > 0$) if evacuee quartile j is greater (less) than the incumbent's quartile k. Finally, we use the results from equation (4) to test the benefits of ability grouping at the school-grade level where students are grouped with other students of the same achievement level. In particular, we test whether a student performs better when surrounded by peers of similar achievement. For ability grouping to be beneficial, it is required that the impact of peers in one's own achievement quartile be greater than the impact from those in any other quartile. Hence, we test whether $\beta_{kk} > \beta_{ki}$ for all $i \neq k$, where k is the incumbent student's quartile. Note that peer effects in these three models may result both from direct peer effects as well as from teacher responses to changes in the composition of students.

III. Peer Effects in Houston

A. *Houston Independent School District Data*

HISD provided us with student-level administrative records from 2002–2003 to 2006–2007, although our regressions start in 2003–2004 to allow controls for lagged outcomes. We use data for all schools within HISD. The data include demographic characteristics, test scores, discipline records, and attendance rates. We use

[24] Duflo, Dupas, and Kremer (2011) develop a nice model of peer effects that includes both teacher responses and direct peer effects and derive propositions of the extreme cases of no teacher responses and no direct peer effects, which they test with their data. Unfortunately, we cannot conduct similar tests of these propositions as these would require us to have information on the previous performance of incumbents and evacuees at the classroom level, and we only have this information at the school and grade levels in Louisiana.

third to tenth grade math and reading/English language arts (ELA) scores from the Texas Assessment of Knowledge and Skills (TAKS) exam, which is the exam used in Texas for accountability purposes. Since some students took multiple exams in a year due to retests but we do not know the dates of the exams, we use each student's minimum score in a year.[25] We standardize the scaled scores using statewide means and standard deviations of scaled scores in each grade and year.[26] More details on the achievement data can be found in our online Appendix. Our discipline data include records for students in grades 1–12 on the number of infractions resulting in an in-school suspension or more severe punishment.[27] Attendance data provides the percent of enrolled days during which a student is present for grades 1–12. In 2005–2006 the data includes 171,659 incumbent and 4,986 evacuee observations in grades 1–12.[28]

Table 1 shows descriptive characteristics for evacuees and incumbents in 2005–2006. Both groups are almost exclusively nonwhite. Evacuees, however, are 90 percent African American while incumbents are 59 percent Hispanic and 29 percent African American. Both incumbents and evacuees are very likely to qualify for free/reduced-price lunch and are very likely to be at-risk, but these characteristics are more prominent among evacuees: 97 percent and 94 percent compared to 79 percent and 69 percent, respectively, for incumbents.[29] Evacuees also have substantially lower achievement than incumbents in both reading and math. For incumbent students, the average fractions of evacuees in their school and their grade-school are both 2.7 percent, with standard deviations of 3.1 and 3.4.[30] Table 2 reports results from regressions of test scores, attendance, and discipline on an indicator of whether the person is an evacuee.[31] The results show that the test scores of Houston evacuees are 0.4 to 1.1 standard deviations lower than incumbents in the same schools and with the same demographic characteristics. Further, evacuees have lower attendance rates.[32] Houston evacuees in middle/high school also tend to have fewer disciplinary

[25] Less than 1 percent of students have multiple math and 4 percent have multiple reading scores in a given year due to retakes. Since not all students have test scores, one may be concerned that this introduces selection bias if the probability of testing is correlated with the entry of evacuees. We test for this by estimating a linear probability model of the likelihood of having being tested on the evacuee share and find a negative effect for both elementary and middle/high, though it is only marginally significant at the 10 percent level for the elementary school sample. We check the robustness of our results to imputing the lagged test score and the average test score in the bottom quartile for those who have missing scores and find them to be robust to these imputations (provided in the online Appendix).

[26] Results with unstandardized scale scores and those standardized within HISD using pre-Katrina scale scores are similar and provided in the online Appendix.

[27] Since evacuee status is identified in 2005–2006 we drop any student in 2006–2007 that was not observed in 2005–2006. Additionally, to ensure that a pre-Katrina lagged score is available we drop grade three in 2005–2006 and grades three and four in 2006–2007.

[28] Online Appendix Table 1 shows the number of observations in each sample overall and by year.

[29] At-risk status is defined as being over-aged for your grade, having a difficult situation at home (e.g., pregnant, foster child) or having low academic performance (below the 40th percentile).

[30] Note that since these are unweighted means, the fraction of Katrina students in the school and grade is higher for evacuees. Online Appendix Table 2 reports school weighted means for all schools and for schools with positive evacuee shares. The school-weighted fractions of evacuees in a school for incumbents and evacuees in the entire sample are 2.4 percent and 2.9 percent. The school-weighted fractions in schools with positive shares of evacuees are the same, 2.9 percent, for both incumbents and evacuees, as expected. This pattern is similar for Louisiana.

[31] Note that the number of observations in this table are different from Table 1 since here we are pooling incumbents and evacuees, but separating elementary from middle and high school students. Also, Table 1 only reports descriptive statistics for 2005–2006, while Table 2 runs regressions for multiple years.

[32] While this difference indicates that absenteeism seemed to be a problem among evacuees, the bulk of evacuees were going to school so that we know they were indeed in contact with incumbent students in schools.

Table 1—Characteristics of Evacuees and Incumbent Louisiana and Houston Students: 2005–2006

	Houston		Louisiana	
	Incumbent	Evacuees	Incumbent	Evacuees
Demographics				
Female	0.490	0.492	0.496	0.499
	(0.500)	(0.500)	(0.500)	(0.500)
White	0.093	0.039	0.544	0.426
	(0.290)	(0.192)	(0.498)	(0.495)
Hispanic	0.585	0.039	0.014	0.016
	(0.493)	(0.193)	(0.119)	(0.126)
African American	0.290	0.903	0.424	0.536
	(0.454)	(0.296)	(0.494)	(0.499)
Asian	0.032	0.019	0.010	0.015
	(0.177)	(0.137)	(0.098)	(0.121)
Free/reduced price lunch	0.787	0.968	0.545	0.762
	(0.409)	(0.175)	(0.498)	(0.426)
At-risk	0.691	0.941	—	—
	(0.462)	(0.235)	—	—
Fraction Katrina/Rita evacuee in school	0.027	0.066	0.032	0.171
	(0.031)	(0.051)	(0.036)	(0.287)
Fraction Katrina/Rita evacuee in grade	0.027	0.073	0.032	0.177
	(0.034)	(0.057)	(0.038)	(0.287)
Observations	171,659	4,986	228,568	8,693
Number of schools	292		1,046	
Number of schools with positive evacuee share	249		798	
Test scores, discipline, and attendance				
Math	−0.493	−1.533	0.000	−0.239
	(1.424)	(1.472)	(1.000)	(1.065)
Observations	112,241	2,151	227,912	8,664
Reading/ELA	−0.686	−1.716	0.000	−0.221
	(1.784)	(1.063)	(1.000)	(1.076)
Observations	113,977	2,307	227,902	8,671
Disciplinary infractions	0.629	0.893	—	—
	(1.698)	(1.987)	—	—
Observations	171,659	4,986	—	—
Attendance rate (percent)	94.55	83.30	—	—
	(8.97)	(18.00)	—	—
Observations	171,659	4,986	—	—

Notes: Standard deviations are provided in parentheses. Incumbent students are restricted to those used in the peer-effects regressions. Hence the Louisiana sample includes grade 4 from 1999–2000 through 2004–2005, grade 5 in 2005–2006, grade 10 in 2001–2002 and 2002–2003, grades 8 and 10 in 2003–2004 and 2004–2005, and grades 6–10 in 2005–2006 and 2006–2007. The sample in Louisiana is also limited to schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. To maintain consistency with later regressions, we restrict to students who have a pre-Katrina outcome measure, with the exception of Houston evacuees since we do not have any pre-Katrina information on those students. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses nonevacuees in schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Disciplinary infractions are the number of suspensions or more severe punishments in a year.

Table 2—Regressions of Test Scores on Katrina/Rita Evacuee Status Dummy

	Houston		Louisiana		
	2005–2006	2006–2007	Lagged outcomes in 2005–2006	2005–2006	2006–2007
Panel A. Elementary					
Math (LEAP and TAKS)	−0.72***	−0.80***	−0.12***	−0.19***	—
	(0.09)	(0.19)	(0.03)	(0.03)	—
Observations	26,282	10,970	36,699	36,688	—
Reading/ELA (LEAP and TAKS)	−0.47***	−0.94***	−0.13***	−0.25***	—
	(0.12)	(0.26)	(0.03)	(0.03)	—
Observations	23,602	8,480	36,701	36,695	—
Disciplinary infractions	−0.04	0.011	—	—	—
	(0.03)	(0.03)	—	—	—
Observations	61,943	40,767	—	—	—
Attendance	−5.95***	−1.66***	—	—	—
	(0.34)	(0.28)	—	—	—
Observations	61,943	40,767	—	—	—
Panel B. Middle/High					
Math (LEAP and TAKS)	−0.64***	−0.43***	−0.09***	−0.14***	−0.11***
	(0.08)	(0.06)	(0.02)	(0.02)	(0.02)
Observations	64,155	56,629	179,822	179,200	174,656
Reading/ELA (LEAP and TAKS)	−1.05***	−0.77***	−0.11***	−0.12***	−0.09***
	(0.14)	(0.11)	(0.02)	(0.02)	(0.02)
Observations	64,871	57,340	179,766	179,119	174,715
Disciplinary infractions	−0.111	0.416***	—	—	—
	(0.069)	(0.132)	—	—	—
Observations	86,078	74,603	—	—	—
Attendance rate (percent)	−13.16***	−4.14***	—	—	—
	(1.17)	(0.65)	—	—	—
Observations	86,078	74,603	—	—	—

Notes: Standard errors are provided in parentheses and clustered by school. Regressions include student's race, gender, free/reduced price lunch status, and grade and school fixed-effects. Incumbent students are restricted to those used in the peer-effects regressions. Hence the Louisiana sample includes grade 4 from 1999–2000 through 2004–2005, grade 5 in 2005–2006, grade 10 in 2001–2002 and 2002–2003, grades 8 and 10 in 2003–2004 and 2004–2005, and grades 6–10 in 2005–2006 and 2006–2007. The sample in Louisiana is also limited to schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. To maintain consistency with later regressions we restrict to students that have a pre-Katrina outcome measure, with the exception of Houston evacuees since we do not have any pre-Katrina information on those students. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses nonevacuees in schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. Houston includes grades 1–12 for discipline and attendance and grades 3–11 for achievement. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Lagged scores are a student's most recent pre-Katrina exam score. Disciplinary infractions are the number of suspensions or more severe punishments in a year.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 3—Reduced-Form Estimates of Evacuee Share of Enrollment in Grade on Incumbent Test Scores: Houston

All (1)	African American (2)	Boys (3)	Girls (4)	Bottom quartile (5)	2nd quartile (6)	3rd quartile (7)	Top quartile (8)	Placebo test (9)
Panel A. Elementary								
Math								
−0.29	−0.83	−0.14	−0.41	−0.73	−1.15**	−0.99*	0.73	−0.16
(0.41)	(0.55)	(0.48)	(0.41)	(0.85)	(0.51)	(0.58)	(0.50)	(0.89)
90,267	23,966	44,868	45,399	20,229	23,183	23,368	23,487	53,255
Reading/ELA								
−0.12	−0.73	−0.30	0.06	−0.42	0.12	0.14	0.18	−0.65
(0.40)	(0.58)	(0.49)	(0.44)	(0.93)	(0.59)	(0.38)	(0.57)	(1.28)
73,923	21,582	37,264	36,659	16,813	19,348	18,977	18,785	42,281
Panel B. Middle/High								
Math								
0.31	−0.05	0.21	0.41	0.20	−0.30	0.05	−0.27	−0.02
(0.35)	(0.48)	(0.39)	(0.35)	(0.85)	(0.39)	(0.30)	(0.37)	(0.45)
249,021	72,314	122,799	126,222	51,011	63,829	66,189	67,992	129,845
Reading/ELA								
0.66	0.55	0.53	0.75*	0.10	0.22	−0.32	−1.07***	−0.30
(0.45)	(0.53)	(0.53)	(0.40)	(1.62)	(0.35)	(0.35)	(0.33)	(0.47)
250,808	72,641	123,639	127,169	50,089	65,660	66,940	68,119	130,310

Notes: Standard errors are provided in parentheses and clustered by school. Observation counts appear in the last row for each subject. Regressions cover 2003–2004 to 2006–2007 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Placebo regressions are limited to pre-Katrina years and apply the 2005–2006 evacuee shares to 2004–2005 observations. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year across the district. Elementary is defined as any student in grades 3–5. Middle/High is any student in grades 6–11.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

infractions than similar incumbents initially and then more in 2006–2007, suggesting schools practiced leniency early on.

B. *Reduced-Form Effects from the Inflow of Evacuees in Houston*

In Table 3 we show the effect of evacuees on math and reading performance of their nonevacuee peers; i.e., the exogenous peer effects. Panels A and B present results for students in elementary grades (3–5) and middle and high grades (6–11), respectively. Column 1 reports estimates for the full sample, while columns 2, 3, and 4 report results of fully saturated models for African Americans, boys and girls. Columns 5–8 report results for incumbents in different parts of the districtwide incumbent achievement distribution based on pre-Katrina scores.

Roughly half of the estimates in Table 3 are negative and most are not statistically significant. This leads us to conclude that the large influx of evacuees (many of whom are low achieving) did not have detectable test score effects on average. The results do show negative and statistically significant effects on math for elementary students in the second and third quartiles. Columns 6 and 7 of panel A show that for elementary students an increase of 10 percentage points in the inflow of evacuees reduces math scores for these nonevacuees by about 0.1 standard deviations. Measuring this effect in the grades and schools that received the highest share of evacuees (i.e., 31 percent) implies that the inflow of evacuees reduced average test scores of incumbent peers in the second and third quartiles by 0.31 standard deviations.[33]

Finally, in column 9 of Table 3 we present results from a falsification test in the spirit of Angrist and Krueger (1999) by regressing the pre-Katrina outcomes in that grade and school on the post-Katrina shares of evacuees in that school and grade as if these shares corresponded to 2004–2005.[34] If we were capturing preexisting trends in the grades and schools, then the coefficients in the share of evacuees should be significant. Thus, we set Katrina share to zero in 2003–2004 and to the 2005–2006 value for observations in 2004–2005, then estimate using only data from 2003–2004 and 2004–2005. None of the coefficients from the falsification test are statistically significant, suggesting that Katrina shares are not correlated with preexisting differences in grade-school trends before the actual inflow of the evacuees. While the tests have relatively weak power and we only have two years of pre-Katrina data, these results suggest no self-selection of evacuees into schools with changing quality at the grade-school level.

In online Appendix Tables 4 and 5 we report a large number of robustness checks for elementary and middle/high schools, respectively. First, we report results using classroom level for elementary in Appendix Table 4, which are similar to those in Table 3 but smaller in magnitude. Second, in order to address potential endogenous movement after the initial evacuation, we report second-stage results from a two-stage least squares (2SLS) model where evacuee share is instrumented using the evacuee share on September 13, 2005 and the share of evacuees living in shelters on October 28, 2005.[35] These estimates are also similar to our baseline models. Third, given that incumbent students may have moved in response to the inflow of evacuees, we provide models that instrument evacuee shares in a student's current school with the share of evacuees in his or her 2004–2005 school. The results are again similar to baseline.

Fourth, we exclude schools with Katrina/Rita shares of more than 10 percent and schools with zero shares to check whether outliers are driving the results. Fifth, we estimate an interrupted panel model that leaves out the year prior to the hurricane, 2004–2005, to ensure that we are not capturing sorting of evacuees into schools that were temporarily worsening. Sixth, in an additional specification we

[33] Online Appendix Table 3 shows results of equation (1) using school-level shares instead of grade-level shares that are similar though larger in magnitude.

[34] Since our identification strategy relies on the exogeneity of grade-level evacuee shares to changes in quality in the school, it is useful to examine simple correlations of mean pre-Katrina incumbent scores within a grade and school and an indicator of whether there were any evacuees in a grade and school. These correlations are small and positive for both Houston and Louisiana, never exceeding 0.15 for each test and sample. Conversely, conditional on having any evacuees the correlation with evacuee share in a school grade is negative and again small, never exceeding 0.2.

[35] Since the Reliant Complex and Convention Center were only meant to be very temporary residences, students in these locations moved either to other shelters, apartments, or to hotels/motels within a few weeks of the storms. These students were then assigned to the zoned schools corresponding to their new locations. Hence we exclude these students from the 9/13/05 enrollment instrument. The first-stage results (available by request) show that both the Katrina/Rita share in shelters and the 9/13/05 enrollment are significant at the 1 percent level.

use unstandardized scaled scores rather than standardized scores and in another one we use scale scores standardized using the districtwide mean and standard deviation for the 2004–2005 cohort. Seventh, to account for nonrandom selection into testing conditional on enrollment, we re-estimate our results substituting missing scores with either the student's most recent lagged test score or with the mean achievement of bottom-quartile students. Eighth, we add evacuee counts directly together with a quadratic on enrollment to separate the effect of the evacuees themselves from changes in enrollment due to other reasons. Ninth, we estimate models that add student fixed effects, school-by-year effects, and school-by-grade effects. In almost all of these cases, we find results that are similar to the baseline specification.

In addition to these checks, we provide models where evacuee share is interacted with an indicator for whether the incumbent student is a new entrant and find that new entrants are more susceptible to peer influences by the evacuees, though the effect is also there for the incumbents. Finally, in online Appendix Table 6 we report results by quintile instead of quartiles to check whether the pattern remains the same—as before, the largest negative peer effect on elementary math occurs in the middle of the distribution and the largest negative effect on middle/high school occurs at the top of the distribution.

Since Hanushek, Kain, and Rivkin (2004) find that disruption from new students generates negative peer effects, a concern is that the results could be driven by disruption generated by the inflows and outflows of the evacuees. This is unlikely, however, to be the main externality in our context as the superintendents, principals, and teachers we interviewed pointed out that most of the new students entered at the very beginning of the academic year and thus little material had been covered by the time the evacuees entered schools. Nonetheless, we examine this potential channel by estimating regressions that include the average weekly evacuee share and the standard deviation of weekly enrollment, instrumenting the latter with the standard deviation of weekly evacuee enrollment. If disruption from the inflows and outflows of the evacuees is the only channel driving the results, we would expect the average effect to disappear when the standard deviation is added. Online Appendix Table 7 shows that the results on average peer scores are similar with or without the standard deviation and that the standard deviation itself does not have a significant effect.[36]

C. *Linear-in-Means Models of Peer Effects in Houston*

In this section we estimate standard models of peer effects in which we regress a student's test scores on the concurrent mean test scores of his or her peers as in equation (2). In columns 1 and 2 of Table 4 we report ordinary least squares (OLS) estimates for elementary students in the pre-Katrina and full samples. The results restricted to pre-Katrina years show that in elementary grades, a student's achievement moves almost one for one with mean peer achievement. Column 2 shows that the relationship is smaller for elementary students when using the full sample.

[36] An alternative check is to see if impacts differ by the timing of when evacuees enroll, as entry later in the year would be more disruptive. To check this we estimate models that add measures of evacuee shares arriving in the first and last half of the school year. While estimates become imprecise as relatively few evacuees arrive after the first half of the year, we nonetheless find little to suggest that the impacts differ by when evacuees enter. These results are provided in online Appendix Table 8.

Table 4—Linear-in-Means Models of Peer Effects in Achievement for Houston
(*OLS and 2SLS models using Grade-Level Evacuee Share as an instrument for peer achievement–elementary*)

	OLS–Pre-Katrina only	OLS–all years	2SLS–first stage	2SLS–second stage	OLS model with peer achievement interacted with own quartile			
Dependent variable →	Avg. peer score	Avg. peer score	Evacuee share	Avg. peer score	Avg. peer score × quartile 1	Avg. peer score × quartile 2	Avg. peer score × quartile 3	Avg. peer score × quartile 4
	(1)	(2)	(3)	(4)	(5)			
Math	0.90***	0.79***	−1.61***	0.18	0.82***	0.77***	0.67***	0.58***
	(0.04)	(0.04)	(0.50)	(0.22)	(0.04)	(0.05)	(0.05)	(0.07)
Observations	54,148	92,259	92,259	92,259	90,259			
F-test of joint equality					7.2***			
Reading	0.84***	0.70***	−1.13**	0.11	0.84***	0.68***	0.59***	0.37***
	(0.05)	(0.05)	(0.51)	(0.31)	(0.06)	(0.06)	(0.08)	(0.08)
Observations	43,053	73,918	73,918	73,918	73,918			
F-test of joint equality					10.5***			

	Reduced form with peer achievement interacted with own quartile				2SLS model with peer achievement interacted with own quartile—second stage instrumented with evacuee share × own quartile with lags			
Dependent variable →	Evacuee share × quartile 1	Evacuee share × quartile 2	Evacuee share × quartile 3	Evacuee share × quartile 4	Avg. peer score × quartile 1	Avg. peer score × quartile 2	Avg. peer score × quartile 3	Avg. peer score × quartile 4
	(6)				(7)			
Math	−0.75	−1.16**	−0.99*	0.73	0.43	0.70**	0.62*	−0.52
	(0.85)	(0.51)	(0.57)	(0.50)	(0.43)	(0.30)	(0.32)	(0.46)
Observations	90,259				90,259			
F-test of joint equality	3.1**				1.5			
Reading	−0.42	0.12	0.15	0.18	0.30	−0.13	−0.15	−0.14
	(0.93)	(0.58)	(0.38)	(0.57)	(0.66)	(0.70)	(0.43)	(0.48)
Observations	73,918				73,918			
F-test of joint equality	0.1				0.1			

Notes: Standard errors are provided in parentheses and clustered by school. Regressions cover 2003–2004 and 2006–2007 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. In regressions 5, 6, and 7, all covariates are interacted with the student's quartile to generate fully interacted models. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year across the district. Elementary is defined as any student in grades 3–5. Middle/High is any student in grades 6–11.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Since the OLS models are potentially endogenous, we exploit the exogenous allocation of evacuees to schools conditional on school fixed-effects and lagged achievement and estimate regressions that use the evacuee share in the grade and school as an instrument for the average quality of peers in a grade-school. Columns 3 and 4 report first- and second-stage estimates of these regressions. Column 3 shows that this is a strong instrument for the elementary school sample.[37] The 2SLS estimates for elementary in column 4 are statistically insignificant and considerably smaller than those in columns 1 and 2, falling to 0.2 standard deviations for math and 0.1 for reading.

While we do not find statistically significant peer effects on average for the non-evacuee student population, we explore whether peer effects differ depending on baseline achievement by estimating equation (3). Column 5 reports the OLS results of the interaction of average peer score with each of the four incumbent quartiles, and column 7 reports the equivalent 2SLS results where the instruments are the evacuee shares interacted with the incumbent quartile indicators. Column 6 is the reduced form. The OLS results are positive for all quartiles in reading and math, with larger peer effects at the bottom and top of the distributions. The 2SLS results, however, only show a large positive effect of peers on math test scores of elementary students in the middle two quintiles. The results are very similar when we exclude lagged test scores and the interactions of lagged test scores with the incumbent's lagged quintile, as shown in online Appendix Table 10.

In Table 4 we report F-statistics of test of joint equality of peer effect coefficients for all quartiles. In the reduced form we reject equality of the coefficients across quartiles of own achievement in math. Given our large standard errors in the 2SLS estimation, however, we cannot reject equality of the peer effects across own quartiles despite the large differences in point estimates. Hence, we cannot reject the null of the linear-in-means model. Unfortunately, the Houston data does not provide us with the ability to investigate this further. Below, we provide evidence against the linear-in-means models using data from Louisiana.

IV. Peer Effects in Louisiana

A. *Louisiana Department of Education Data*

Louisiana testing data comes from the Department of Education's Division of Standards, Assessment, and Accountability via Data Recognition Corporation and covers all students in the state who took the state criterion-referenced Louisiana Educational Assessment Program (LEAP/iLEAP) exams from the 1999–2000 to 2006–2007 academic years. The data are at the student level and include information on gender, race/ethnicity, free/reduced-price lunch status, and achievement. Test scores are available for grades three through ten after Katrina, but only for grades four, eight, and ten before Katrina in math as well as English and language arts.[38] We can only estimate value-added specifications for middle/high school students

[37] Since the instrument is uniformly weak for the middle/high school sample, we do not report these results here and instead provide them in online Appendix Table 9.

[38] Science and social studies exams are also available in some years. We follow the existing education literature and focus on the math and reading/language exams.

who have a pre-Katrina lagged score in fourth or eighth grade and thus we limit that sample to 2001–2002 and later.[39] We also provide results for elementary school students without lagged test scores.[40] More details on the LEAP/iLEAP exams can be found in the online Appendix.

We exclude schools in the parishes directly affected by Hurricanes Katrina and Rita[41] and all schools with more than 70 percent evacuees in 2005–2006, since these also tended to be in heavily affected areas. We further exclude observations in 2006–2007 for students who are not observed in 2005–2006 as their evacuee status is unknown. This leaves us with 536,009 math and 522,392 ELA incumbent observations in total for middle/high school and 287,181 math and 287,230 ELA observations for elementary school (see online Appendix Table 1).

Table 1 shows descriptive statistics in 2005–2006 for the 8,693 evacuees and 228,568 nonevacuees in our final sample. It is noteworthy that 54 percent of evacuees in Louisiana are African American compared to 42 percent of nonevacuees. Also, evacuees are more economically disadvantaged with 76 percent eligible for free or reduced-price lunch versus only 54 percent of incumbents.

Test scores are measured as standard deviations within a grade and year, including all incumbent test-takers after making the sample restrictions above. We only use nonevacuees in the standardization procedure to ensure that our estimates are relative to the average student who was not directly affected by the storms. Table 1 shows that Louisiana evacuees score about one-quarter of a standard deviation below nonevacuee students in 2005–2006. Table 2 reports differences in test scores after controlling for individual characteristics and school effects so that we can compare evacuees to incumbents in the same post-Katrina schools. These results show that pre-Katrina math and ELA scores of evacuees (lagged scores in 2005–2006) in primary schooling are 0.12 and 0.13 standard deviations lower than those of nonevacuees, and these differences persist post-Katrina. In middle and high school, prestorm scores of evacuees are 0.09 and 0.11 standard deviations lower than those of nonevacuees, again with poststorm scores showing similar differences. These means hide substantial variation in evacuee achievement, however. Online Appendix Figure 4 shows the position of evacuees relative to incumbents in their receiving schools in 2004–2005. Evacuees are more likely to be in the first three deciles of their new school compared to their incumbent counterparts, while they are less likely to be in deciles four through ten, though there remain many evacuees at the upper end of the distribution. This suggests that incumbents were often exposed to both higher- and lower-achieving peers.

[39] This grade structure limits us to students in grade ten from 2001–2002 through 2002–2003, grades eight and ten from 2003–2004 through 2004–2005, and grades six through ten in later years.

[40] Since the data includes only students who take the exams, one may be concerned about differential attrition from the tested sample with the inflow of evacuees. To check whether this is an important concern, we collected enrollment data and calculated the impact of the inflow of evacuees from each achievement quartile on the ratio of tested to enrolled students by grade and school. With the exception of third-quartile evacuees in math and second quartile evacuees in ELA, we find no significant effect. These results are reported in online Appendix Table 11.

[41] These include the parishes of Orleans, Jefferson, Saint Bernard, and Plaquemines, which were affected by Hurricane Katrina, and the parishes of Cameron and Calcasieu, which were affected by Hurricane Rita.

Table 5—Reduced-Form Estimates of Evacuee Share of Enrollment in Grade on Incumbent Test Scores: Louisiana

	All	African American	Boys	Girls	Bottom quartile	2nd quartile	3rd quartile	Top quartile	Placebo test
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A. Elementary (no lags)									
Math									
	0.54***	0.38	0.40*	0.72***	−0.14	0.04	0.31	0.47*	−0.11
	(0.18)	(0.27)	(0.22)	(0.21)	(0.32)	(0.23)	(0.25)	(0.28)	(0.20)
	287,181	129,579	147,813	139,368	65,025	71,606	74,138	76,412	184,201
ELA									
	−0.02	−0.26*	−0.03	−0.02	−0.48	−0.20	−0.31	−0.18	−0.26*
	(0.18)	(0.15)	(0.26)	(0.18)	(0.34)	(0.20)	(0.21)	(0.23)	(0.15)
	287,230	229,678	147,842	139,388	65,210	72,013	73,885	76,122	184,246
Panel B. Middle/High (with lags)									
Math									
	−0.13	−0.06	−0.11	−0.15	−0.17	−0.16	0.01	−0.02	0.05
	(0.11)	(0.28)	(0.12)	(0.12)	(0.17)	(0.14)	(0.12)	(0.21)	(0.12)
	536,009	129,602	263,450	272,559	123,316	133,757	137,800	141,136	132,486
ELA									
	−0.03	−0.04	−0.11	0.04	0.32	−0.01	−0.00	−0.12	−0.49
	(0.10)	(0.16)	(0.11)	(0.10)	(0.20)	(0.13)	(0.13)	(0.16)	(0.37)
	522,392	218,930	259,140	263,252	116,304	128,894	135,911	141,283	122,847

Notes: Standard errors are provided in parentheses and clustered by school. Observation counts appear in the last row for each subject. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses nonevacuees in schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Placebo regressions are limited to pre-Katrina years and apply the 2006–2007 and 2007–2008 evacuee shares to 2003–2004 and 2004–2005 observations, respectively. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999–2000 through 2004–2005 and grade 5 in 2005–2006. Middle/High includes grade 10 in 2001–2002 and 2002–2003, grades 8 and 10 in 2003–2004 and 2004–2005, and grades 6–10 in 2005–2006 and 2006–2007.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

B. *Reduced-Form Effects of the Inflow of Evacuees in Louisiana*

We begin by estimating equation (1). Table 5 shows these results. Overall there is little evidence that the presence of evacuees has a negative effect *on average*. Elementary students actually show gains in math scores associated with increases in the share of evacuees in their grade. The effects for ELA scores are small and negative for all elementary students, though more negative and marginally statistically significant for African American students. Panel B in Table 5 reports reduced-form effects of the impact of evacuees on incumbent test scores for middle and high school students. We show results for the whole sample in column 1 and we separate by race, gender, and prior test score quartile in columns 2–8. Thirteen of the 16 coefficients in these columns are negative, though most are modest in size and none are statistically significant. Online Appendix Tables 12 and 13 shows results estimated

Table 6—Linear-in-Means Models of Peer Effects in Achievement for Louisiana
(*OLS and 2SLS models using evacuee share in grade × mean evacuee lagged achievement as an instrument for peer achievement*)

	OLS—Pre-Katrina only	OLS—all years	2SLS—first stage	2SLS—second stage	OLS model with peer achievement interacted with own quartile			
	Avg. peer score	Avg. peer score	Evac. share × mean lagged evac. score	Avg. peer score	Avg. peer score × quartile 1	Avg. peer score × quartile 2	Avg. peer score × quartile 3	Avg. peer score × quartile 4
	(1)	(2)	(3)	(4)	(5)			
Panel A. Elementary (no lags)								
Math	0.56***	0.58***	1.69***	0.33**	0.29***	0.32***	0.33***	0.40***
	(0.02)	(0.02)	(0.38)	(0.15)	(0.02)	(0.02)	(0.02)	(0.02)
Observations	251,763	287,090	287,090	287,090	287,090			
F-test of joint equality					5.2**			
ELA	0.44***	0.46***	1.07***	−0.00	0.29***	0.25***	0.24***	0.25***
	(0.03)	(0.03)	(0.36)	(0.27)	(0.03)	(0.02)	(0.03)	(0.02)
Observations	251,805	287,139	287,139	287,139	287,139			
F-test of joint equality					0.9			
Panel B. Middle/High (with lags)								
Math	0.31***	0.34***	2.55***	0.15*	0.27***	0.29***	0.26***	0.41***
	(0.04)	(0.06)	(0.82)	(0.08)	(0.03)	(0.04)	(0.09)	(0.14)
Observations	192,843	535,302	535,302	535,302	535,302			
F-test of joint equality					3.7**			
ELA	0.25***	0.41***	1.91***	0.08	0.42***	0.41***	0.32***	0.28***
	(0.02)	(0.02)	(0.43)	(0.08)	(0.02)	(0.02)	(0.02)	(0.04)
Observations	179,217	520,569	520,569	520,569	520,569			
F-test of joint equality					13.9***			

(*Continued*)

using school-level shares. For elementary school the results are similar to Table 5. For middle/high school the results remain negative and are bigger in magnitude, as in Houston, with 9 out of 16 coefficients significant.

C. *Linear-in-Means Models of Peer Effects in Louisiana*

After estimating reduced-form regressions, we proceed to estimate equation (2) for elementary and middle/high school students in Louisiana. Columns 1 and 2 in Table 6 report the OLS estimates for the pre-Katrina and full samples. Columns 3 and 4 report the first and second stages of our instrumental variables specification. We use the achievement-adjusted evacuee share in the school and grade, i.e., the evacuee share multiplied by the mean pre-Katrina achievement of the evacuees, to instrument for mean peer achievement. Thus, the identification assumption here is

Table 6—Linear-in-Means Models of Peer Effects in Achievement for Louisiana
(*OLS and 2SLS models using evacuee share in grade x mean evacuee lagged achievement as an instrument for peer achievement*) (*Continued*)

	Reduced-form model with peer achievement interacted with own quartile				2SLS model with peer achievement interacted with own quartile—second stage; instrumented with evacuee share × mean lagged evacuee score × own quartile			
	Avg. evacuee score × evacuee share × quartile 1	Avg. evacuee score × evacuee share × quartile 2	Avg. evacuee score × evacuee share × quartile 3	Avg. evacuee score × evacuee share × quartile 4	Avg. peer score × quartile 1	Avg. peer score × quartile 2	Avg. peer score × quartile 3	Avg. peer score × quartile 4
	(6)				(7)			
Panel A. Elementary (*no lags*)								
Math	1.14**	0.96***	1.99***	1.57***	0.80***	0.62***	1.00***	0.76***
	(0.47)	(0.34)	(0.42)	(0.55)	(0.27)	(0.16)	(0.15)	(0.19)
Observations	287,090				287,090			
F-test of joint equality	2.7**				1.4			
ELA	1.12*	−0.07	0.81**	0.49	0.83**	−0.09	0.72***	0.47
	(0.60)	(0.31)	(0.41)	(0.33)	(0.36)	(0.39)	(0.25)	(0.29)
Observations	287,139				287,139			
F-test of joint equality	2.3*				1.3			
Panel B. Middle/High (*with lags*)								
Math	0.69***	1.19***	0.77***	0.88***	0.35***	0.64***	0.23*	0.31
	(0.23)	(0.21)	(0.22)	(0.31)	(0.10)	(0.22)	(0.13)	(0.21)
Observations	535,302				535,302			
F-test of joint equality	2.9**				4.6***			
ELA	0.61**	0.42*	1.00***	0.45*	0.38***	0.21*	0.58***	0.23
	(0.27)	(0.23)	(0.21)	(0.26)	(0.12)	(0.11)	(0.18)	(0.20)
Observations	520,569				520,569			
F-test of joint equality	3.8**				2.6*			

Notes: Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses nonevacuees in schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Placebo regressions are limited to pre-Katrina years and apply the 2006–2007 and 2007–2008 evacuee shares to 2003–2004 and 2004–2005 observations, respectively. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. In regressions 5, 6, and 7 all covariates are interacted with the student's quartile to generate fully interacted models. Middle/High includes grade 10 in 2001–2002 and 2002–2003, grades 8 and 10 in 2003–2004 and 2004–2005, and grades 6–10 in 2005–2006 and 2006–2007.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

that the quantity and quality of the evacuees in a given grade-school is not related to changes in quality at the school level. For elementary students pre-Katrina, for whom we cannot control for lagged achievement, the OLS results suggest that a one standard deviation increase in the achievement of peers correlates with 0.56 and 0.44 standard deviation increases in own math and ELA scores, respectively. The comparable coefficients for middle and high school students are 0.31 and 0.25 standard deviation increases in own math and ELA scores. The relation is even stronger when we include the post-Katrina years.

Column 3 shows that there is a strong first-stage relationship between the achievement-adjusted evacuee share and the average peer achievement. The second stage results in column 4 show no significant effect of peer quality on ELA test scores, but a positive and significant effect on math test scores. A one standard deviation increase in all peers' test scores increases math scores by 0.33 standard deviations for elementary students and 0.15 standard deviations for middle and high school students.

Columns 5 and 7 provide OLS and 2SLS estimates of equation (3) that allow peer effects to differ by incumbent achievement levels. The OLS estimates show positive, large, and statistically significant peer effects for students in all quartiles. The 2SLS estimates show moderately larger and still statistically significant effects. For middle and high school students, the test of equality of all coefficients for math is rejected at the 1 percent level. We also reject equality of coefficients for ELA for middle and high school students at the 10 percent level. Results of models without lagged test scores and without interactions of the lagged test scores with the quartile dummies reported in online Appendix Table 14 show similar rejection of equality of the math and ELA coefficients for middle/high school. For elementary students we cannot reject the null of linear in means for both math and ELA, but these results do not control for lagged test scores. Hence, these results suggest that linear-in-means may not be the best model for explaining peer effects, at least in secondary grades. Thus, in the next section we explore models that allow for nonlinearities in peer effects.

D. *Nonlinear Models and the Structure of Peer Effects*

In this section we estimate equation (4), which separates evacuee share by pre-Katrina achievement quartiles. We estimate models where all incumbents are pooled as well as fully saturated models of equation (4) split by the incumbent students' pre-Katrina achievement quartiles. These specifications, which can only be estimated for Louisiana where we have evacuees' pre-Katrina performance measures, allow us to examine how impacts vary by both evacuee and incumbent achievement.

Panels A and B of Table 7 report results from these specifications for math and ELA scores, respectively. Column 1 shows the impact of evacuees in different quartiles of the pre-Katrina achievement distribution on all incumbent students. These results show a negative and significant effect of Katrina evacuees in the bottom quartile on incumbent students in math. By contrast, evacuees in the third and top quartiles have a positive but insignificant effect on incumbents. For ELA there is a negative but not significant impact from bottom-quartile evacuees and a positive but not significant impact from top-quartile evacuees.

When we estimate fully saturated models in columns 2 through 5, both the negative effects from bottom-quartile evacuees and the positive effects from

Table 7—Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement: Louisiana Middle/High

Pre-Katrina LEAP quartile	All (1)	Bottom quartile (2)	2nd quartile (3)	3rd quartile (4)	Top quartile (5)
Panel A. Math					
Katrina/Rita share in quartile 1	−0.79***	−0.68**	−1.56***	−1.31***	−1.67***
	(0.25)	(0.31)	(0.29)	(0.38)	(0.54)
Katrina/Rita share in quartile 2	−0.23	−0.56	−0.41	−0.11	0.02
	(0.31)	(0.45)	(0.40)	(0.38)	(0.61)
Katrina/Rita share in quartile 3	0.34	0.32	0.69*	0.73*	0.48
	(0.31)	(0.47)	(0.40)	(0.39)	(0.51)
Katrina/Rita share in quartile 4	0.42	1.25**	1.24***	0.75*	0.93*
	(0.31)	(0.54)	(0.41)	(0.39)	(0.56)
Observations	536,009	123,316	133,757	137,800	141,136
Estimate of Q4–Q1	1.21***	1.93***	2.79***	2.06***	2.59***
	(0.44)	(0.66)	(0.54)	(0.60)	(0.83)
Panel B. ELA					
Katrina/Rita share in quartile 1	−0.16	−0.44	−0.45	−1.02***	−0.53
	(0.28)	(0.35)	(0.38)	(0.35)	(0.44)
Katrina/Rita share in quartile 2	0.03	0.74	−0.18	−0.40	−0.51
	(0.35)	(0.57)	(0.48)	(0.44)	(0.42)
Katrina/Rita share in quartile 3	−0.23	0.18	−0.03	−0.13	−0.03
	(0.35)	(0.58)	(0.48)	(0.42)	(0.47)
Katrina/Rita share in quartile 4	0.37	1.62**	1.11**	1.81***	0.46
	(0.27)	(0.65)	(0.45)	(0.36)	(0.40)
Observations	522,392	116,304	128,894	135,911	141,283
Estimate of Q4–Q1	0.54	2.06***	1.56**	2.83***	0.98
	(0.42)	(0.77)	(0.66)	(0.53)	(0.64)

Notes: Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses nonevacuees in schools where there are < 70 percent evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999–2000 through 2004–2005 and grade 5 in 2005–2006. Middle/High includes grade 10 in 2001–2002 and 2002–2003, grades 8 and 10 in 2003–2004 and 2004–2005, and grades 6–10 in 2005–2006 and 2006–2007.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

top-quartile evacuees are large and significant for math and ELA. For example, a 10 percentage point increase in bottom-quartile evacuees reduces math achievement of top-quartile incumbents by 0.17 standard deviations while the impact on bottom-quartile incumbents is smaller at 0.07 standard deviations. On the other hand, a similar increase in top-quartile evacuees increases incumbent scores by 0.13 in the bottom quartile and by 0.09 in the top quartile. The impact on ELA is similar; a 10 percentage point increase in bottom quartile evacuees reduces incumbent achievement by 0.04 to 0.10 standard deviations. A similar increase in top-quartile evacuees increases incumbent achievement by 0.05 to 0.18 standard deviations. Given that a 10 percentage point shift at the top or bottom of the achievement distribution is a large change in peer composition, these estimates are of plausible magnitudes. We also provide estimates of the differential impacts of receiving top or bottom-quartile evacuees. The estimates suggest that an increase of 10 percentage points in top-quartile evacuees increases achievement by 0.10 to 0.28 standard deviations more than an equivalent increase in bottom-quartile evacuees, depending on the incumbent's own quartile and the subject.

Online Appendix Tables 15 and 16 show similar patterns using school-level or test administrator (our proxy for classroom)–level shares.[42] We find consistently larger and more significant effects in these nonlinear models when we use school-level fraction Katrina. This may indicate that there are school-level spillovers or cross-grade peer interactions that are important in determining own test scores. When we move to classroom-level fraction Katrina, we see the same patterns, though somewhat smaller and more precisely estimated effects, in which the addition of low-scoring evacuees harms incumbent students while the addition of high-scoring evacuees helps incumbents.

These specifications also allow us to test a number of hypotheses and models proposed in the peer effects literature. We test three different channels through which peer effects may operate. First, we test a monotonic model in which high-achieving peers help everyone and low-achieving peers hurt everyone. This could be the result of direct peer-to-peer learning if high-achieving peers teach others in the classroom and low-achieving peers are disruptive. This could also result from teacher responses, however, if teachers raise the target level of learning or put more effort into teaching when high achievers enter the class and slow down the pace of teaching when low achievers enter. We test a weak form of this in which we ask whether the effect from evacuees at the top quartile is statistically significantly greater than the effect of evacuees from the bottom quartile. Our results are consistent with this hypothesis with or without a Bonferroni correction. We also test a strong version of this model by testing whether the effect from evacuees in quartile j is significantly higher than the effect from evacuees in quartile $j - 1$.

The results of these and other tests are reported in Table 8, assuming the tests are independent. Specifically, we count the number of tests that are significant at the 10 percent level in the direction predicted by the various models, insignificant, or significant in the opposite direction predicted by the model. For the weak form of monotonicity for math scores, four of four tests are significant and in the direction predicted. For ELA, three tests are significant and in the predicted direction. Our tests for the strong form of monotonicity (every coefficient statistically significantly greater than the preceding quartile) are less conclusive. In online Appendix Table 20, we report similar results but allowing for a Bonferroni adjustment that allows for interdependence between tests. Using the Bonferroni adjustment we get similar conclusions—weak monotonicity is accepted four out of four times for math and three out of four times for ELA.

Next, we test whether direct peer effects could be working through invidious comparisons with peers. In particular, if a student is grouped with peers under his

[42] Since evacuees were more likely to have endogenous classroom assignments in the second year, we conduct test administrator–level specifications that instrument for 2006–2007 evacuee shares with a student's 2005–2006 shares. These are provided in online Appendix Table 17 and provide similar results. Online Appendix Table 18 reports results of equation (4) for all quartiles but without lags for both the full and lag samples. The results are consistent but smaller than what we find with lags. Finally, online Appendix Table 19 reports results for elementary, which cannot be estimated with lagged test scores but similarly show negative effects from the inflow of lower quartile evacuees and positive effects from the inflow of higher quartile evacuees.

Table 8—Tests of Models of Peer Effects Using Estimates from Table 7

	Math				ELA			
Model	Number of estimates significant in direction the model suggests	Number of estimates insignificant	Number of estimates significant in opposite direction	Consistent, inconsistent, or neither?	Number of estimates significant in direction the model suggests	Number of estimates insignificant	Number of estimates significant in opposite direction	Consistent, inconsistent, or neither?
(1) Weak monotonicity: Effect of evacuees from top quartile is statistically significantly greater than effect from bottom quartile.	4	0	0	Consistent	3	1	0	Consistent
(2) Strong monotonicity: Effect of evacuees from quartile j is statistically significantly greater than effect from quartile $j-1$.	4	8	0	Consistent	1	11	0	Neither
(3) Invidious comparison: Effect of evacuees from lower quartiles than the incumbent is positive and effect from evacuees in higher quartiles is negative.	0	6	6	Inconsistent	0	8	4	Inconsistent
(4) Ability grouping: Effect of evacuees from own quartile is statistically significantly greater than effect from higher quartiles.	0	2	4	Inconsistent	0	3	3	Inconsistent

Note: See text for the specific tests conducted for each model.

level, his self-confidence and performance may improve, with the opposite case if he is grouped with higher-achieving peers. We reject this model with or without Bonferroni corrections. For example, for math scores, 6 of 12 tests are significant in the opposite direction, while the other 6 tests are statistically insignificant.

Finally, we test whether students benefit from being grouped with other students of the same level. Hence, we test whether the impact of an increase in evacuees from the incumbent student's same quartile is greater than the impact from higher-achieving evacuees. The focus on comparing the effect of similar peers to higher achievers and not lower achievers allows us to distinguish this model from monotonicity. While grouping is thought to be useful because teachers can adjust their teaching to a more appropriate level for the students, we find no evidence that grouping students at the school-grade level by ability is beneficial for low-achieving students.

Hence, we conclude that the best model for describing our results is a nonlinear model with monotonic peer effects. Moreover, results from the nonlinear models show patterns that are more consistent with peer effects than with alternative stories of the impact of Katrina and Rita evacuees on schools. For example, if the inflow of evacuees reduced school resources (either by increasing class size, reducing expenditures per student, or reducing the quality of teachers), both high-achieving and low-achieving evacuees should reduce incumbent performance. Instead, we find positive impacts from high achievers and negative from low achievers.

E. *Robustness Checks: Nonlinear Models*

The placebo tests for middle/high in Section IVA, which control for pre-Katrina lagged test scores, show no evidence of the overall Katrina share being related to preexisting trends in Louisiana. It may be, however, that the quality of Katrina evacuees is correlated with preexisting trends. For our nonlinear results to be capturing trends, it would have to be that high- and low-quality evacuees selected into schools with improving and worsening trends, respectively. We conduct a few robustness checks to address this potential issue. First, we estimate equation (2) including school-specific linear trends (off of the entire data and off of the pre-Katrina data alone), separate trends for schools with above- or below-median evacuee shares, or separate trends for schools with above- or below-median free/reduced-price lunch rates. In all of these cases, with results provided in online Appendix Tables 21–24, we find estimates that are similar to those in Table 7, although the results with school-specific time trends are somewhat weaker and less precise. Hence, our results are robust to controls for differential trends.

Second, we conduct a placebo test using school-level data that regresses the pre-Katrina School Performance Score (SPS), a composite measure used to determine accountability ratings, on future overall Katrina shares and future Katrina shares in different quartiles in Table 9.[43] In particular, using data only from 1999–2000 through 2004–2005, we estimate difference-in-differences models where we set the evacuee share variable in a given year to be equal to the actual evacuee share two years (column 1) or three years (column 2) later. We then regress the SPS on the placebo evacuee shares, year dummies, school fixed-effects, and school-grade means of the covariates used in our main model.[44] Only the estimate on quartile 1 evacuees in column 1 is significant and only at the 10 percent level. To further check for selective trends, in online Appendix Table 26 we provide estimates of the future (2005–2006) evacuee shares on school fixed effects, year dummies, and the interaction of evacuee share with year indicators. This allows us to test whether changes in SPS are related to future evacuee inflow. None of the coefficients that interact the year indicators with the overall share are significant and only 1 estimate out of 20 that interacts year indicators with evacuee shares by quartile is significant. If evacuees of each achievement level are differentially selecting into schools with trends, we would expect to see consistently significant estimates from these regressions. Nonetheless, since the single significant coefficient comes from 2004–2005, we further report estimates from interrupted panel versions of the nonlinear models that leave out 2004–2005 in online Appendix Table 27 and these show the same monotonic patterns and similar effects to those discussed above.[45]

[43] The SPS is based 90 percent on an assessment index and 10 percent on an attendance index for K–5 schools; 90 percent on an assessment index, 5 percent on an attendance index, and 5 percent on a dropout index for K–8 and 7–8 schools; and 70 percent on an assessment index and 30 percent on a graduation index in 9–12 schools.

[44] See the notes to Table 9 for a full list of controls. In online Appendix Table 25 we provide the same analysis with no controls beyond school and year fixed effects. The results are similar except the marginally significant estimate on quartile 1 evacuees in the 2-year (column 1) model becomes significant at the 5 percent level and increases to 56.

[45] We also note that the significant estimate for quartile 1 evacuees in Table 9 appears to be driven by observations in 2004–2005, hence the interrupted panel also helps allay concerns that there may be some correlation with evacuee share in that quartile and school quality changes in 2004–2005.

Table 9—School-Level Placebo Test Regressions of School Performance Score on Evacuee Shares in Louisiana: 1999–2000 through 2004–2005

	School performance scores in year t regressed on evacuee shares in year $t + 2$ (1)	School performance scores in year t regressed on evacuee shares in year $t + 3$ (2)
Katrina/Rita share	−3.7	−2.1
	(3.3)	(2.8)
Observations	5,738	4,795
Katrina/Rita share in quartile 1	−38.1*	−23.6
	(20.8)	(23.9)
Katrina/Rita share in quartile 2	−29.4	1.8
	(32.9)	(27.4)
Katrina/Rita share in quartile 3	25.2	9.9
	(27.3)	(29.2)
Katrina/Rita share in quartile 4	14.5	−0.9
	(21.2)	(19.2)
Observations	5,738	4,795
Mean School Performance Score	83.1	
Standard Deviation of SPS	20.9	

Notes: Robust standard errors clustered by school are provided in parentheses and clustered by school. Regressions also include school fixed-effects, year dummies, percent of tested students who are African American, Hispanic, Asian, free/reduced-price lunch, and the distribution of tested students across grades in each year. School performance score (SPS) is a metric used to determine accountability ratings that includes test scores, attendance rates, dropouts, and graduation rates. See text for more details. In column 1, evacuee shares in 2005–2006 and 2006–2007 are matched to the same schools in 2003–2004 and 2004–2005, respectively, so that all SPS scores in the regression are pre-Katrina. For 1999–2000 through 2002–2003 evacuee shares are set to zero. For column 2 the analysis is conducted one year earlier. Quartiles for evacuees in Louisiana are within grade across the state data and determined from their most recent pre-Katrina score. Since SPS is an aggregate measure that includes both math and reading scores, evacuee shares are averaged across math and reading quartiles.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

We conduct additional robustness checks similar to those done for the Houston results. First, we check whether excluding schools with more than 10 percent of evacuee shares and limiting to schools with positive shares changes the results. Second, we use unstandardized scaled scores and alternative standardization using only pre-Katrina test scores instead of standardized test scores by grade and year. Third, to deal with potential movements of incumbent students, we use the evacuee share in the incumbent's student most recent pre-Katrina school to instrument for the current evacuee share. Fourth, we add student fixed effects or school-grade fixed effects to the regressions. Fifth, we check whether the results are similar if we do not interact lagged achievement with the number of years between tests and whether including counts of evacuees and controlling for a quartic in enrollment separately changes the results. Next, we control for the percent black and percent of students with free lunch to make sure we are not simply capturing the effects of changes in composition. After, we separate by quintiles instead of quartiles to see if the results are robust to this change. Then, we check robustness to standardization using pre-Katrina data only. Finally, we check if the results hold if we limit our checks to schools that get a positive evacuee share. These results can be found in online Appendix Tables 28–39. In all of these cases the results are similar to those found in Table 7.[46]

Hence, overall we find evidence that peer effects are important but smaller than suggested by OLS estimates. Moreover, we reject linear-in-means models or peer effects about half the time (more than half the time if we use school- rather than grade-level Katrina shares) and provide evidence that the peer effects are structured nonlinearly and monotonically. Finally, our results do not support invidious comparison or ability grouping (at the school-grade level) models of peer effects.

V. Effects of Evacuees on Incumbent Students' Attendance and Behavior

Our detailed administrative data also allow an examination of the effects of peers on attendance and discipline. Aside from the impact that students may have on others' academic performance, they may affect behavior and willingness to accept and follow rules. Our interviews with principals and teachers in Houston indicated that even basic rules such as showing up to school on time or at all were problematic with some of the evacuees. Likewise, our interview of the Superintendent of the West Baton Rouge Parish school district told us that they "had problems in schools with turf issues … [though these were] mostly hallway and recess things." News reports and our own interviews pointed to behavioral problems related to the evacuees. For example, some elementary teachers indicated that evacuees were more likely to "talk back to the teachers" and that some of the nonevacuees imitated this behavior. In secondary grades, the differences in behavior between evacuee and nonevacuee students manifested more in terms of truancy, fighting, and engaging in risky behaviors.

Panels A and B of Table 10 present reduced-form effects of evacuees on incumbent attendance rates in Houston from estimating equation (1) for elementary and middle/high school. While the results show no effects on elementary students, there is an increase in absenteeism in secondary schools. An increase in evacuee share of 10 percentage points reduces attendance by 0.4 percentage points, though this is not statistically significant.[47] Attendance was particularly problematic after the arrival of evacuees for African American incumbent students, with a 10 percentage point inflow of Katrina students generating a reduction in the attendance rate of 0.6 percentage points in secondary schools. We note, however, that these effects, while significant, are relatively small. In a 180-day school year on average, a 10 percentage point increase in evacuee share in these grades increases African American absenteeism by close to 1 day. An impact of this size is unlikely to generate a detectable effect on test scores. While there is a lack of evidence of the impact of attendance on achievement, Marcotte and Hemelt (2008) find that for eighth-grade students, school closures only start to significantly affect test scores after ten days. Thus, we

[46] We also estimate regressions with school-year fixed-effects which allow us to rely exclusively on within-year variation in evacuee share across grades. While the same basic pattern emerges for math, the results become mainly insignificant. For ELA, the monotonic patterns go away. These results are provided in online Appendix Table 41. While arguably more likely to be exogenous, however, the intraschool variation is much smaller than the variation we use in our main estimates, hence we believe these estimates reflect only very small differences in peers and are not informative as to what happens when there are more substantial changes.

[47] We provide placebo results as in Table 3 that are also insignificant.

Table 10—Estimates of Evacuee Share of Enrollment in Grade on Incumbent Attendance and Disciplinary Infractions: Houston

	All (1)	African American (2)	Boys (5)	Girls (6)	Placebo test (7)
Panel A. Elementary					
Attendance rate (percent)	−0.20	−0.54	−0.78	1.28	−1.11
	(0.72)	(1.53)	(0.80)	(1.04)	(0.69)
Observations	223,776	60,710	115,104	108,672	123,366
Disciplinary infractions	−0.05	−0.26	−0.01	−0.12	−0.25
	(0.15)	(0.32)	(0.25)	(0.11)	(0.35)
Observations	223,776	60,710	115,104	108,672	123,366
Panel B. Middle/High					
Attendance rate (percent)	−3.9	−6.3**	−5.2	−2.6	−5.6
	(3.4)	(3.1)	(4.2)	(2.8)	(3.5)
Observations	323,499	98,595	163,133	160,366	166,202
Disciplinary infractions	1.86	2.73*	2.18	1.43*	−1.12
	(1.18)	(1.44)	(1.57)	(0.84)	(1.62)
Observations	323,499	98,595	163,133	160,366	166,202

Notes: Standard errors are provided in parentheses and clustered by school. Disciplinary infractions are the number of times a student receives an in-school suspension or more severe punishement in a year. Regressions cover 2003–2004 and 2006–2007 and include student's race, gender, free lunch/reduced price lunch status, school fixed-effects, grade-by-year fixed-effects, and the lagged dependent variable from the student's most recent pre-Katrina lagged dependent variable interacted with the number of years between exams. The placebo test in column 7 includes 2003–2004 and 2004–2005 only and apply 2005–2006 Katrina/Rita share to 2004–2005 observations. Elementary is defined as any student in grades 1–5. Middle/High is any student in grades 6–12.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

believe that the effects on test scores are working through the peer effects channels described above rather than through absenteeism.

Table 10 also shows results from regressing disciplinary infractions on Katrina shares using equation (1). We note that these estimates should be interpreted with caution, as they could result from changes in enforcement as well as behavior. Nonetheless, our discussions with school officials suggest that if anything, enforcement was more lenient, generating a downward bias. The results in panel B show an increase in disciplinary infractions for African Americans and girls in secondary schools in Houston.[48] A 10 percentage point increase in the inflow of Katrina students raised disciplinary infractions by about 0.4 and 0.2 among African Americans and girls, respectively.[49]

[48] As with achievement, we report school-level results in online Appendix Table 40; specification tests are in online Appendix Table 41, and of tests of disruption effects in online Appendix Table 42. In general, the results are similar to the baseline results.

[49] While we have discipline data for Louisiana, the earliest year is 2004–2005 and hence we forgo impact models as any pre-Katrina observation will not be able to include a lag. Nonetheless, we can use this data to test whether pre-Katrina evacuee discipline has an impact on our achievement estimates. In online Appendix Table 43 we add the share of evacuees with a disciplinary infraction in 2004–2005 to the models in Table 7. The estimates are almost all statistically insignificant and induce little change in the estimates of evacuee share on achievement by quartile.

VI. Alternative Effects of Katrina Evacuees and External Validity

One interpretation of the results above is that they are due to peer effects. Another possibility, however, is that the inflow of evacuees reduced resources or incumbent students left the schools that received evacuees. To test these alternatives, we conduct analyses of how evacuee share affects class size, expenditures, teacher quality, and school switching. In online Appendix Table 45 we report results of school-level regressions of class sizes on the share of evacuees in various quartiles in Louisiana.[50] These results show no statistically significant relation between class sizes and the evacuee shares by quartiles for both elementary and secondary schools, with the exception of a marginally significant decline in elementary classes of size 21–26 for quartile 1. Online Appendix Table 46 shows results of regressions of operating and instructional expenditures per student, teacher characteristics, and average class sizes on the evacuee share in Houston.[51] We find no statistically significant effect on per-student operating or instructional expenditures, class sizes, teacher experience, teacher education, or certification in secondary schools. In elementary schools, certification and graduate degrees are significantly *higher* at the 10 percent level although the magnitude of the effect is economically small. For example, for an average elementary school with 35 teachers, a 10 percentage point increase in evacuee share would add only one additional teacher with a graduate degree.[52] We also note that there is little evidence that graduate education and certification increase achievement.[53] Additionally, this result would, if anything, bias us against finding peer effects in elementary students in Houston, which is actually the group of students with the largest effects.

We estimate student-level models for Houston where the dependent variable is whether the student switches schools or leaves HISD the following year.[54] These results are provided in online Appendix Table 46. We find no statistically significant change in the mobility of students between HISD schools.[55] Similarly, we find no effect on mobility out of the district for secondary students in Houston. We do find a *reduction* in district leavers at the elementary level, but it is only marginally significant. When we look at whether incumbent students in different quartiles respond to the inflow of Katrina evacuees, we find that elementary students in the top quartile are more likely to leave the district and middle/high school students at the bottom quartile are more likely to change schools. Assuming that better students within each quartile are more likely to leave, however, these results should

[50] Instead of information on average class size, Louisiana only provides information on the percent of classes in a school with 1–20, 21–26, and 27 or more students.

[51] The expenditure and teacher quality variables are not available for Louisiana. Since we only have information at the school level and schools span several grades, we cannot control for grade effects. Instead, we control for percent in each grade and interact with year dummies.

[52] This is not surprising as an HISD official told us that the district targeted teachers at job fairs and hired a combination of retired, substitute, and displaced teachers. In Louisiana most of the newly hired teachers were displaced teachers from the New Orleans area.

[53] For example, Angrist and Guryan (2008) find no evidence that certification requirements increase the quality of teachers and presumably performance.

[54] To avoid capturing switches and departures due to normal progression to middle and high school and due to graduation, we limit the school switching estimates to students who are not in the maximum grade for their school. We also limit the district leaver regressions to students in grades 1–11.

[55] Since we only have grades four, eight, and ten prior to Katrina, we cannot estimate switching or leaving regressions in Louisiana.

bias us toward finding negative effects for those at the top quartile in elementary school and for those in the bottom quartile in middle/high school. Table 3 suggests larger negative effects at the middle of the distribution in elementary and at the top end of the distribution in high school.

Overall, we find little evidence that the effects from the inflow of evacuees operate through reduced resources or incumbent flight. Another concern, however, is that the inflow of Katrina evacuees may be working through neighborhood effects. Unfortunately, we cannot examine this because our data does not identify students' neighborhoods. This would, however, be part of a peer effect working at a different level. Moreover, these effects are more likely to exhibit in estimates based on school-level evacuee shares than grade-level shares. The fact that our estimates are smaller at the grade level suggests that they purge a substantial portion of any neighborhood effects. A more important concern is whether the effects we find are simply capturing preexisting neighborhood conditions. While we cannot rule out this possibility, the neighborhood effects would have to be strongly correlated with lagged evacuee achievement within school and grade in order to explain our results.

Finally, an important concern is how much we can learn about peer effects more generally from our study. Clearly, there were special circumstances surrounding the hurricanes. While we have tried to rule out alternative channels through which the arrival of the evacuees may have affected student performance, the nature of the evacuations leaves the possibility that our results could be in part due to the unique circumstances surrounding this event. Nonetheless, we note that any situation with large changes in peers will be unique and one must weigh the benefits of having sizable exogenous variation and external validity.[56] In our view the former is more important, but we acknowledge that others may disagree.

VII. Conclusion

In this paper we examine the impact of an exogenous inflow of students with a wide range of achievement levels on the academic performance and behaviors of their peers by examining the effects of the absorption of evacuees from Hurricanes Katrina and Rita into the Houston Independent School District and into the non-evacuated school districts in Louisiana.

We use student-level data from HISD and from Louisiana's Department of Education to estimate the impact on math and language test scores, disciplinary infractions, and absenteeism. We find that, on average, evacuees had no impacts on incumbent achievement in Houston, with the exception of effects on elementary students in the middle of the distribution. Similarly, we find little evidence of reduced-form impacts of evacuees on incumbents in Louisiana. We find evidence of negative effects on attendance and

[56] Another issue one may worry about is the possibility of Hawthorne effects; if there were a superhuman effort on the part of many to minimize the damage done to both evacuees and incumbent students that could not be maintained in a more general context. While we cannot rule this out, we note that if this were the case we would expect that in other circumstances peer effects from low achievers would have an even larger negative effect while there should be little difference in peer effects from high achievers. On the other hand, if teachers made greater than usual efforts to focus attention on the evacuees, then this would lead us to find less negative effects on incumbents under normal circumstances. Our interviews suggest, however, that while teachers may have put additional effort into helping evacuees by putting in extra hours, this did not come at the expense of incumbents during the regular school day.

behavior for middle/high school students in Houston, however, with larger effects on African Americans. We try to get inside the peer-effects black box by testing a variety of peer-effects models. We reject linear-in-means models of peer effects and find evidence in support of monotonic peer effects, where students benefit from high-achieving peers and are harmed by low-achieving peers, in secondary grades. These results suggest that peer effects work through students learning from one another or from teacher responses to changes in the student distribution. By contrast, our results are inconsistent with invidious comparison and ability grouping models. We also find little to suggest that these results are driven by disruption.

While it is difficult to think of any event that caused as large and exogenous a change in peer composition as Hurricanes Katrina and Rita, admittedly our results could be explained by other mechanisms. Nonetheless, we find little to suggest that evacuees affected class size, expenditures per student, teacher quality, or student mobility, so we interpret our findings as capturing peer effects. In addition, the finding that evacuees of various achievement levels affecting incumbents differently is consistent with peer effects driving our results and is inconsistent with alternative explanations such as neighborhood effects that should affect all students in the same manner.

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